Math 1232-20: Single-Variable Calculus II

GWU Summer 2026 Session II

Meetings:	MTWR, 4:00 - 5:30 Corcoran Hall 103 06/29/2026 - 08/08/2026	**Textbook:**	OpenStax Calculus Vol. 1 OpenStax Calculus Vol. 2 by Strang and Herman
Instructor: **Email:**	Ben Clingenpeel ben.clingenpeel@gwu.edu	**Office:** **Office hours:**	Phillips 720G MTR 2:00-3:30
Course page:	https://bncling.github.io/courses/math-1232-2026		

Course content and learning outcomes

Though condensed into just six weeks, this course covers the material of a second semester of a standard year-long sequence in single-variable calculus: the calculus of exponential and logarithmic functions, L'Hôpital's rule, techniques of integration, infinite series and Taylor series, and polar coordinates

By the end of the course, students will acquire the following skills and knowledge. Students will be able to:

- define logarithm, exponential, and inverse trigonometric functions, explain their basic properties (continuity, derivatives, asymptotes, etc.) and recognize their graphs;
- apply these functions to word problems, and correctly interpret the results;
- solve integrals using integration by parts, trigonometric substitution and partial fractions;
- determine whether an infinite series converges or diverges;
- express algebraic and transcendental functions using Maclaurin and Taylor series; and
- analyze, create and recognize polar and parametric graphs.

Course structure

The course will be run on the 'flipped' model. This means (paraphrasing a more precise definition from Talbert's *Flipped Learning* [4]) that we are moving the first contact with new material out of the traditional classroom lecture and into a more flexible, "individual learning space." In our class meetings, we will be actively practicing this new material together. That is to say, I will be posting the lectures I previously would have given in class online for you to watch at home before class, and then during class you'll do worksheets in groups, with problems I would previously have assigned as homework.

Why do this? In short, I think this will help you learn the material better. Learning is an active process! You can't learn to play an instrument or a sport just by watching someone else do it, and similarly, the hardest part of learning something like Calc II is getting the necessary practice. A more traditional lecture-in-class structure has you doing this more difficult work at home, but this is the time when you stand to benefit the most from my support. So there will still be some practice I assign outside of the classroom (see the description of the WeBWorK system below), but I plan to have you do a significant chunk of it in class because that's where your peers and I can answer questions as they come up. Moreover, moving the lectures into a format where you can pause and rewind also means you have more control over the pace at which the new material is delivered.

Does research support this? Yes! A meta-analysis of studies on active learning—an underlying principle of the flipped model—finds that students in STEM courses with an active learning approach scored about a half of a standard deviation better than their peers in more traditional lecture courses on average, and that the effect size increased in small courses (like ours!) [1]. They also find that the failure rate in the more traditional courses was over 50% higher than the failure rate in the active learning courses! A more recent meta-analysis on flipped learning gives a similar half of a standard deviation performance improvement for the students in flipped classrooms relative to students in unflipped classrooms [3], and a meta-analysis just of studies on flipped math courses also finds a significant positive effect, albeit slightly more moderate [2]. That said, they found that in courses utilizing a "structured formative assessment" at the beginning of class (I'll be calling these *lecture check-ins*), there was an average performance increase exceeding the results of [1] and [3].

[1] Scott Freeman, Sarah L Eddy, Miles McDonough, Michelle K Smith, Nnadozie Okoroafor, Hannah Jordt, and Mary Pat Wenderoth. Active learning increases student performance in science, engineering, and mathematics. *Proceedings of the national academy of sciences*, 111(23):8410–8415, 2014.

[2] Chung Kwan Lo, Khe Foon Hew, and Gaowei Chen. Toward a set of design principles for mathematics flipped classrooms: A synthesis of research in mathematics education. *Educational Research Review*, 22:50–73, 2017.

[3] Peter Strelan, Amanda Osborn, and Edward Palmer. The flipped classroom: A meta-analysis of effects on student performance across disciplines and education levels. *Educational Research Review*, 30:100314, 2020.

[4] Robert Talbert and Jon Bergmann. *Flipped learning: A guide for higher education faculty*. Routledge, 2023.

What will a typical class look like? Class will begin with a short lecture check-in, so it is important that you arrive on time. Afterwards, I will give a *very* brief overview of the concepts covered in the lecture, focusing on the questions you have for me and on whatever you tell me you'd like to hear about the most in your lecture check-ins. The bulk of the class time will be spent on solving problems, initially individually and then in groups, and I will be wandering through the room to see how things are going, answer questions, and offer hints as appropriate. I will collect worksheets at the end of the day and post some additional practice problems through WeBWorK. At the end of each week we will have a quiz (weeks 1, 2, 4, and 5), or an exam (weeks 3 and 6). See below for the precise weighting of these components, as well as for a more detailed description of each.

Grading system. Your grade will be based on the following:

- Lecture check-ins – 5%
- In-class work – 15%
- Online homework through WeBWorK – 10%
- Quizzes – 15%
- A midterm exam – 25%
- A final exam – 30%

Minimum scores for each letter grade are as follows: A, 93%; A-, 90%; B+, 87%; B, 83%; B-, 80%; C+, 77%; C, 73%; C-, 70%; D+, 67%; D, 63%; D-, 60%.

Lecture check-ins. These will be brief, open-note quizzes emphasizing terminology and definitions. (I might ask you to define something or to give an example of something meeting a definition from the lectures.) One of the main points of doing these will be for me to get a sense of what you have and have not understood from the lectures, and so on the lecture check-in I will also explicitly ask you to tell me what you've found most confusing or would most like to hear about briefly before we start the main in-class assignment.

In-class work. The majority of class time will be spent on a worksheet of several problems of varying difficulty:

- *Warm-up problems.* If you've worked through the lectures, these should be straightforward problems that are quick to solve. We'll start working on the worksheets individually, ideally with enough time that you would feel confident presenting your solution to any of these warm-up problems with your group members, as this will be the first thing you do once we start the group work. If you have more time, you should absolutely look ahead at the next set of problems!

- *Medium problems.* These should mostly still be straightforward, though should take you a bit longer and will benefit a bit more from collaboration. These problems are at the same level as the problems you will see on quizzes and exams, so working through them is good preparation. After presenting solutions to the warm-up problems, you should work through these medium problems as a group, making sure everyone is on the same page. This may mean slowing yourself down and explaining some ideas to your peers! See more about this in the section on how to work in groups below.

- *Stretch problems.* These won't always be hard problems per se, but will look a little different from the more calculation focused warm-up and medium problems.

As you work on these, I will be around to help point you in the right direction, checking in periodically. Please call me over if you reach a point where you are collectively stuck! *Although these problems are to be solved as a group, every group member must write their own solutions.* I will collect worksheets at the end of class, and will try my best to pass them back to you at the beginning of the next meeting. These will be graded mostly on completion, and since I expect the warm-up and medium problems to take most of the time, a successful answer to a stretch problem will be strictly for bonus points. That is, there will not be any penalty if you don't have time to think about them, although I do of course encourage you to think about them more on your own later. I will also drop at least the lowest two worksheet scores.

WeBWorK. Accessed through Blackboard under the Assignments tab, WeBWorK is an online homework system. The point of WeBWorK is to give you some additional problems to practice with. You will get as many attempts as you need to answer all WeBWorK problems, but there is a due date for each problem set, usually a few days after the set becomes available. If you need an extension, there is a reduced scoring period available where you can still earn 75% credit. A given problem set will open the on the same day we discuss the material in class, and be due two class meetings later at the end of the day, which a reduced scoring period available through an additional two class periods. Thus a problem set that opens on a Tuesday will be due at 11:59pm on Thursday, with a reduced scoring period open until 11:59pm on the following Tuesday. Ideally, you should think of the reduced scoring period as a last resort. Staying on top of the WeBWorK is one of the best ways to help yourself through the course.

Quizzes. We will have four of these, tentatively scheduled for July 2, July 9, July 23, and July 30. These will be short (20min) quizzes on problems just like the ones you'll encounter in the 'medium' category on the group worksheets. If for some acceptable reason you need to miss a quiz, your quiz grade will be based on the other three quizzes (missing for religious holidays is an exception).

Exams. The midterm will be July 16, and the final will be August 6. As with the quizzes, the problems will be in the 'medium' category. I will post some mock exams before each so you have an idea of what to expect as far as the format, number of questions, etc. If for some acceptable reason you cannot take the midterm, I will adjust the weight of the final exam to cover both the midterm and the final (55%), and again missing for a religious holiday is an exception here.

General course policies, advice, musings

How to work in groups. Since much of the course meeting time will be spent working with your peers, I think it's worth stating the obvious: you should endeavor to treat everyone with respect! This means that if you find yourself in a position of relatively good understanding, you should be willing to take the time to help your group members work through a problem, even if it prevents you from having time in class to work on the more challenging problems (I encourage you to think about them later, if you're interested). Similarly, being a respectful group member also means being willing to volunteer an idea, even if you are not sure it's really on the right track. The best way to improve your understanding is to try something and see what happens, and the goal is that you will all serve as sounding boards for one another.

Classroom technology and AI tools. *This will be a tech-free classroom.* That is, I am requiring that you keep your phones, laptops, tablets, etc. put away for the duration of our class meeting time. The goal of the course structure is to get you engaging with the material directly through active work in groups, so I don't want you to be distracted during this time. I also want to encourage you to avoid outsourcing any of this necessary practice to AI, and since I don't want to have to police whether you are using phones, laptops, or tablets appropriately or not, I think it's best to keep them tucked away. Since I do want you to be able to bring your notes from the lectures with you to class, *a notebook and pencil are required.* The one exception to this policy is that you are allowed a four-function calculator (not a scientific or graphic calculator). That said, I will not write problems that require much arithmetic—no calculator will be necessary. If you have questions about whether a specific calculator you have is acceptable, please ask!

Tips on watching the lectures. It can be easy to watch lectures (especially pre-recorded lectures) very passively, so I urge you to combat this by actively taking notes, just as you would in a usual lecture. Having notes you can bring to class are especially important given the technology policy—I may put a relevant formula or two up on the board, but it is your responsibility to come to class prepared to discuss the material, and this includes having any notes that will help you solve problems. As an aside, you are now deep into the syllabus, and if you're reading this, you've earned bonus points on the final exam. To redeem them, email me a picture of a pet or a favorite animal. Anyhow, a benefit of having pre-recorded lectures is that you can watch them at your own pace, and engage with them a bit more actively. For instance, When I work an example problem, I *highly* encourage you to pause the video and give the example a try on your own! I hope you'll find the ability to re-watch portions of the lecture helpful as well.

Textbook and additional Resources

Course textbook: Our primary text is OpenStax Calculus Volume 2 by Gilbert Strang and Edwin Herman. Some topics often covered in Calculus I at other institutions are deferred to Calculus II at GW, and so we will begin the course using our secondary textbook, OpenStax Calculus Volume 1, also by Gilbert Strang and Edwin Herman. Both texts can be read online for free at the OpenStax website. It's always a good idea to get even more practice than what I'll be assigning through the in-class work and the WeBWorK, and the textbook is a great source of problems.

Peer tutoring: GW offers a peer tutoring service through the Academic Commons. You can book a completely free 50 minute tutoring session here: https://academiccommons.gwu.edu/tutoring.

Mental health services: The University's Mental Health Services offers 24/7 assistance and referral to address students' personal, social, career, and study skills problems. Services for students include: crisis and emergency mental health consultations confidential assessment, counseling services (individual and small group), and referrals. For additional information see:
https://counselingcenter.gwu.edu/ or call 202-994-5300.

Religious holidays and other excused absences

In accordance with University policy, students should notify faculty during the first week of the semester of their intention to be absent from class on their day(s) of religious observance. For details and policy, see "Religious Holidays" at the Office of the Provost at https://provost.gwu.edu/policies-procedures-and-guidelines.

Accommodations

Any student who may need an accommodation based on the impact of a disability should contact the Office of Disability Support Services (DSS) to inquire about the documentation necessary to establish eligibility and to coordinate a plan of reasonable and appropriate accommodations. DSS is located in Rome Hall, Suite 102. For additional information, please call DSS at 202-994-8250, or consult https://disabilitysupport.gwu.edu.

Academic integrity

The complete GWU academic integrity code can be viewed at https://studentconduct.gwu.edu. Per the code, "academic dishonesty is defined as cheating of any kind, including misrepresenting one's own work, taking credit for the work of others without crediting them and without appropriate authorization, and the fabrication of information." One thing to highlight here is that the definition given of cheating includes "engaging in unauthorized collaboration in any academic exercise." Much of the work you'll be doing in this course is collaborative by design, and you also have my permission to collaborate on the WeBWorK assignments, though as with the in-class worksheets, you need to make sure you understand everything your collaborators do.

Safety and security

In the case of an emergency, if at all possible, the class should shelter in place. If the building that the class is in is affected, follow the evacuation procedures for the building. After evacuation, seek shelter at a predetermined rendezvous location. Review the Emergency Response Handbook at https://safety.gwu.edu/emergency-response-handbook.

Schedule

This is an overview of the topics we'll be covering, along with when I think we'll cover things (but this is subject to change). Red cells below are exams and blue cells are quizzes.

June 29	Introduction to the course	July 20	Sequences
June 30	Inverse functions and their derivatives	July 21	Series, divergence and integral tests
July 1	Exponential and logarithmic functions, differentiation	July 22	Comparison tests
July 2	Inverse trigonometric functions, integration	July 23	Alternating series test
July 6	L'Hôpital's rule	July 27	Ratio and root tests
July 7	Integrating products of functions	July 28	Power series and their properties
July 8	Trigonometric substitution	July 29	Taylor series
July 9	Partial fractions	July 30	Working with Taylor series
July 13	Improper integrals	Aug 3	Parametric equations, calculus with parametrics
July 14	Arc length, surface area	Aug 4	Polar coordinates, area and arc length of polar curves
July 15	Introduction to differential equations	Aug 5	Review day
July 16	Midterm exam	Aug 6	Final exam